An infinite maximal tree would give an infinite descending sequence of ordinals

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Editorial note. The counted unit is the article's own Lemma lem:sads-wf-forcing-wf-jump at source lines 1250-1259 together with the proof environment that begins two lines after the statement ends (1261-1292); the pairing is the ordinary adjacent one. This label uses the author's plain form rather than a typed one, and the gate recognises both.

Editorial note. Eight context spans are retained: the Shore-blocking lemma at 426-431, the definitions of a condition, of a split pair and of the two forcing relations at 1114, 1117-1119 and 1121-1127, the definitions of trees and satisfaction at 1169-1183 and 1186-1192, and the statements of the two SADS tree lemmas at 1194-1196 and 1206-1215. All are prerequisites that the counted statement and proof use; their own proofs are not retained and they are counted by sibling cards of this lane.

Editorial note. The retained preamble reproduces verbatim the author's document class at source line 2, his theorem declarations at source lines 16-25, his macro definitions at source lines 46-88 and 91, and his natural-operation macros at source lines 104-107, and it retains mathabx and cleveref; the stix-dependent LS2 block and the inputenc load with the utf8x option are not retained. mathabx supplies the natural sum and product symbols used in the rank definition.

Editorial note. No retained line contains a citation, so the card carries no bibliography.

Editorial note. The two remarks that follow the proof at source lines 1294-1302 are not counted here and are not used as premises.

Allowed standard prerequisites (editorial).

- The two formulas Phi_k and Psi_k and the Shore-blocking functional. Phi_k(G,x) states that there are indices below k, increasing in number x, at which the functional converges on the generic; Psi_k(G,x) states that the Shore-blocking functional Gamma_k converges at x, where Gamma_k is the functional of the Shore-blocking lemma for the ordinal alpha; the author's own definitions are retained at source lines 1250-1251 and the underlying lemma at 426-431.. They are the two formulas out of which

the counted lemma lets the pair be chosen, and the quantities f_0 and f_1 in the proof are built from their values.

- Maximal trees of split pairs and their finite-tree forcing consequence. For every pair of Sigma^0_1 formulas and every condition there is a maximal tree of split pairs satisfying the pair, and if that tree is M-finite then some extension forces the two formulas at two indices and forces the negation of one of the next instances; retained with this card at source lines 1194-1196 and 1206-1215.. The proof obtains the maximal tree from the first statement and concludes with the second once it proves that the tree is M-finite, which is the only remaining step.

- Ordinal-valued rank of a finite tree and well-foundedness of the supremum. To each finite tree one assigns the ordinal obtained by summing, over the nodes, the ordinal omega raised to the sum of the two values f_0 and f_1 at the node, multiplied by the number of missing immediate successors; the hypothesis WF of the supremum of omega to the natural product with omega below alpha forbids an M-infinite decreasing sequence of such ordinals; the article uses these ordinals in the proof.. The proof defines exactly such a rank zeta_s for each approximating tree and shows that the ranks strictly decrease, which contradicts the well-foundedness hypothesis.

Lemma 0.1. *Consider a model $\mathcal{M} = (M, S)$ and $\alpha \in M$ an ordinal less than ϵ_0. For every $k \in M$, there is a Turing functional Γ_k such that, letting $\beta \in \alpha$ be the largest ordinal less than α of index at most k, for every $X \in 2^M$ such that $\mathcal{M}[X] \models \mathsf{RCA}_0$, if there is some $e < k$ such that Φ_e^X is an M-infinite decreasing sequence of elements smaller than β, then Γ_k^X is an M-infinite decreasing sequence of elements smaller than $\beta \dot{\times} k$.*

Moreover, an index of Γ_k can be found computably in k.

A condition is a pair (σ, τ) with σ an M-finite L-ascending sequence included in U and τ an M-finite L-descending sequence included in V. We have $(\sigma_2, \tau_2) \leqslant (\sigma_1, \tau_1)$ if $\sigma_2 \succeq \sigma_1$ and $\tau_2 \succeq \tau_1$.

Definition 0.2. *A* split pair *is a pair (σ, τ) of strings such that σ is an M-finite L-ascending sequence, τ is an M-finite L-descending sequence, and $\max_L \sigma <_L \min_L \tau$.*

Definition 0.3. *Let (σ_0, σ_1) be a condition, $e, x \in M$ and $i < 2$, we write:*

- $(\sigma_0, \sigma_1) \Vdash_i \Phi_e^{G \oplus Y}(x) \downarrow$ *if* $\Phi_e^{\sigma_i \oplus Y}(x) \downarrow$.
- $(\sigma_0, \sigma_1) \Vdash_i \Phi_e^{G \oplus Y}(x) \uparrow$ *if* $\forall \sigma_i' \succeq \sigma_i$ *such that* $(\sigma_i', \sigma_{1-i})$ *is a split pair,* $\Phi_e^{\sigma_i' \oplus Y}(x) \uparrow$.

Definition 0.4. *Fix a condition (σ, τ).*

A (σ, τ)-tree *is a binary tree T labelled by a family of split pairs $(\sigma_\rho, \tau_\rho)_{\rho \in T}$ such that for all $\rho \in T$:*

- $\sigma \preceq \sigma_\rho$ *and* $\tau \preceq \tau_\rho$ *;*
- *For every* $\mu \in T$*, if* $\rho \cdot 0 \preceq \mu$ *then* $\sigma_\rho \preceq \sigma_\mu$ *and if* $\rho \cdot 1 \preceq \mu$ *then* $\tau_\rho \preceq \tau_\mu$ *;*

A node ρ *is said to be* valid *if for every* $\mu \in T$ *such that* $\mu \cdot 0 \preceq \rho$*,* $\sigma_\mu \subseteq U$ *and for every* $\mu \in T$ *such that* $\mu \cdot 1 \preceq \rho$*,* $\tau_\mu \subseteq V$*. Being valid is a* $\Delta^0_2(Y)$ *condition.*

Definition 0.5. *Let* T *a* (σ, τ)*-tree*

- *A node* (σ_ρ, τ_ρ) *in* T satisfies *a pair of* $\Sigma^0_1(Y)$*-formulas* $(\varphi_0(G_0, x), \varphi_1(G_1, x))$ *if* $\varphi_0(\sigma, |\rho| - \operatorname{card} \rho)$ *and* $\varphi_1(\tau, \operatorname{card} \rho)$ *holds (where* $\operatorname{card} \rho$ *is the cardinal of* ρ *seen as a set).*
- T satisfies $(\varphi_0(G_0, x), \varphi_1(G_1, x))$ *if every node of* T *satisfies it.*

Lemma 0.6. *For every pair* $(\varphi_0(G_0, x), \varphi_1(G_1, x))$ *of* $\Sigma^0_1(Y)$ *formulas and every condition* (σ, τ)*, there is a maximal* Σ^0_1 (σ, τ)*-tree* T *satisfying it.*

Lemma 0.7. *Let* (σ, τ) *be a condition,* $(\varphi_0(G_0, x), \varphi_1(G_1, x))$ *be a pair of* $\Sigma^0_1(Y)$ *formulas, and* T *be an* M*-finite maximal* (σ, τ)*-tree satisfying* $(\varphi_0(G_0, x), \varphi_1(G_1, x))$*. Then there is an extension* $(\hat{\sigma}, \hat{\tau}) \Subset (\sigma, \tau)$ *and some* $p, q \in M \cup \{-1\}$ *such that the following holds:*

- $(\hat{\sigma}, \hat{\tau}) \Vdash_0 \varphi_0(G_0, p)$ *and* $(\hat{\sigma}, \hat{\tau}) \Vdash_1 \varphi_1(G_1, q)$
- $(\hat{\sigma}, \hat{\tau}) \Vdash_0 \neg\varphi_0(G_1, p + 1)$ *or* $(\hat{\sigma}, \hat{\tau}) \Vdash_1 \neg\varphi_1(G_1, q + 1)$*.*

With the convention that $(\hat{\sigma}, \hat{\tau}) \Vdash_i \varphi_i(G^i, -1)$ *always holds.*

Lemma 0.8. *Let* $k \in M$*, let* $\Phi_k(G, x)$ *be the formula:* $\exists e_0 < e_1 < \cdots < e_x < k$ *such that* $\Phi^{G \oplus Y}_{e_i}(e_i) \downarrow$ *for all* $i < x$*, let* $\Psi_k(G, x)$ *be the formula:* $\Gamma^{G \oplus Y}_k(x) \downarrow$ *(with* Γ_k *the functional of Lemma 0.1 for the order* α*).*

For any combination $\phi_0(G, x), \phi_1(G, x) \in \{\Phi_k(G, x), \Psi_k(G, x)\}$ *and every condition* (σ, τ)*, there exists an extension* $(\hat{\sigma}, \hat{\tau}) \Subset (\sigma, \tau)$ *and some* $p, q \in M \cup \{-1\}$ *such that the following holds:*

- $(\hat{\sigma}, \hat{\tau}) \Vdash_0 \varphi_0(G_0, p)$ *and* $(\hat{\sigma}, \hat{\tau}) \Vdash_1 \varphi_1(G_1, q)$
- $(\hat{\sigma}, \hat{\tau}) \Vdash_0 \neg\varphi_0(G_1, p + 1)$ *or* $(\hat{\sigma}, \hat{\tau}) \Vdash_1 \neg\varphi_1(G_1, q + 1)$*.*

With the convention that $(\hat{\sigma}, \hat{\tau}) \Vdash_i \varphi_i(G_i, -1)$ *always holds.*

Proof. By Lemma 0.6 there exists a maximal Σ^0_1 (σ, τ)-tree T satisfying

$$(\varphi_0(G, x), \varphi_1(G, x)).$$

To apply Lemma 0.7 and conclude, we just need to prove the M-finiteness of T.

By twisting Γ_k, we can assume that for every $e, n \in M$, if $\Gamma^{\sigma \oplus Y}_k(n) \downarrow$, then

- $\Gamma^{\sigma \oplus Y}_k(m) \downarrow$ for every $m < n$;
- $\Gamma^{\sigma \oplus Y}_k(0), \Gamma^{\sigma \oplus Y}_k(1), \ldots, \Gamma^{\sigma \oplus Y}_k(m)$ is a strictly $<_{\epsilon_0}$-decreasing sequence of elements smaller than $\beta \dot{\times} k$.

Assume by contradiction that T is M-infinite and let $(T_s)_{s\in M}$ be an increasing sequence of computable tree approaching T. Without loss of generality we can assume that $T_s \neq T_{s+1}$ for all $s \in M$.

For every $s \in M$ and ρ node of T_s indexed by (σ_ρ, τ_ρ) with set of immediate successors S_ρ ($\operatorname{card} S_\rho \leqslant 2$ since T_s is a binary tree), define inductively :

$$\zeta_s(\rho) = \omega^{f_0(\rho)\dot{+}f_1(\rho)}(2 - \operatorname{card} S_\rho) \ \dot{+} \sum\nolimits^{\cdot}_{\mu\in S_\rho} \zeta_s(\mu)$$

Where $f_0(\rho) = \Gamma_k^{\sigma_\rho\oplus Y}(|\rho| - \operatorname{card}\rho)$ if $\varphi_0(G,x) = \Psi_k(G,x)$ and $f_0(\rho) = k - (|\rho| - \operatorname{card}\rho)$ if $\varphi_0(G,x) = \Phi_k(G,x)$. And $f_1(\rho) = \Gamma_k^{\tau_\rho\oplus Y}(\operatorname{card}\rho)$ if $\varphi_1(G,x) = \Psi_k(G,x)$ and $f_1(\rho) = k - \operatorname{card}\rho$ if $\varphi_1(G,x) = \Phi_k(G,x)$.

And let $\zeta_s = \zeta_s(\epsilon)$.

Claim. *The sequence ζ_s is an M-infinitely $<_{\epsilon_0}$-decreasing sequence of ordinals less than $\omega^{\beta\times\omega}$*

Let $s \in M$, since $T_{s+1} \neq T_s$, there is a $\rho \in T_s$ an $i < 2$ such that $T_{s+1} = T_s \cup \rho\cdot i$. Therefore:

$$\zeta_s(\rho) \dot{+} (\omega^{f_0(\rho\cdot i)+f_1(\rho\cdot i)} \dot{\times} 2) = \zeta_{s+1}(\rho) \dot{+} \omega^{f_0(\rho)\dot{+}f_1(\rho)}$$

But for every possible values of φ_0 and φ_1, we have $f_0(\rho) > f_0(\rho\cdot 0)$, $f_0(\rho) = f_0(\rho\cdot 1)$, $f_1(\rho) = f_1(\rho\cdot 0)$ and $f_1(\rho) > f_1(\rho\cdot 1)$. So for every possible value of i, we have

$$\omega^{f_0(\rho)+f_1(\rho)} > \omega^{f_0(\rho\cdot i)+f_1(\rho\cdot i)} \times 2$$

and therefore $\zeta_{s+1}(\rho) < \zeta_s(\rho)$ which gives us $\zeta_{s+1} < \zeta_s$ (as $\rho\cdot i$ is the only difference between T_s and T_{s+1}).

Therefore, for all $s \in M$, $\zeta_s \leqslant_{\epsilon_0} \zeta_0 \leqslant_{\epsilon_0} \omega^{\beta\times\omega}$.

This contradicts $\mathsf{WF}(\sup_{\beta<\alpha}\omega^{\beta\times\omega})$, so T is M-finite and the result follows. □